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Comprehensive, Open-Source, and Automated Workflow for Multisite λ-Dynamics in Lead Optimization

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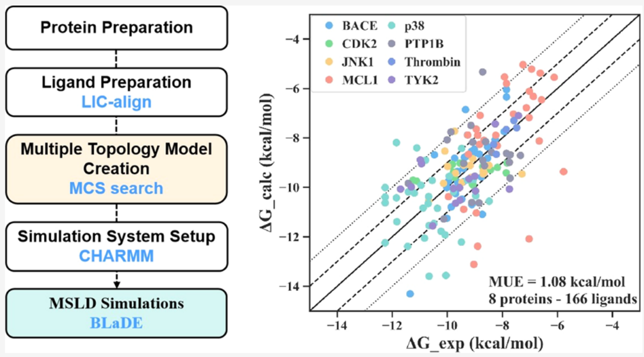
ABSTRACT: Multisite λ-dynamics (MSLD) is a highly efficient binding free energy calculation method that samples multiple ligands in a single round by assigning different λ values to the alchemical part of each ligand. This method holds great promise for lead optimization (LO) in drug discovery. However, the complex data preparation and simulation process limits its widespread application in diverse protein−ligand systems. To address this challenge, we developed a comprehensive, open-source, and automated workflow for MSLD calculations based on the BLaDE dynamics engine. This workflow incorporates the Ligand Internal and Cartesian coordinate reconstruction-based alignment algorithm (LIC-align) and an optimized maximum common substructure (MCS) search algorithm to accurately generate MSLD multiple topologies with ideal perturbation patterns. Furthermore, our workflow is highly modularized, allowing straightforward integration and extension of various simulation techniques, and is highly accessible to nonexperts. This workflow was validated by calculating the relative binding free energies of large-scale congeneric ligands, many of which have large perturbing groups. The agreement between the calculations and experiments was excellent, with an average unsigned error of 1.08 ± 0.47 kcal/mol. More than 57.1% of the ligands had an error of less than 1.0 kcal/mol, and the perturbations of 6 targets were fully connected via the calculations, while those of 2 targets were connected via both calculations and experimental data. The Pearson correlation coefficient reached 0.88, indicating that the MSLD workflow provides accurate predictions that can guide lead optimization in drug discovery. We also examined the impact of single-site versus multisite perturbations, ligand grouping by perturbing group size, and the position of the anchor atom on the MSLD performance. By integrating our proposed LIC-align and optimized MCS search algorithm along with the coping strategies to handle challenging molecular substructures, our workflow can handle many realistic scenarios more reasonably than all previously published methods. Moreover, we observed that our MSLD workflow achieved similar accuracy to free energy perturbation (FEP) while improving computational efficiency by over 1 order of magnitude in speedup. These findings provide valuable insights and strategies for further MSLD development, making MSLD a competitive tool for lead optimization.



■ INTRODUCTION

Free energy is a fundamental concept in statistical physics, holds the key to a clear comprehension of complex molecular processes in biological systems, and provides clues for rational drug design. Therefore, developing efficient and accurate free energy calculation methods is of the utmost importance in drug discovery.[1,2] Among the numerous scenarios in drug discovery, we focus here on lead optimization (LO), which requires a highly accurate estimation of relative binding free energy (RBFE). A fast and reliable RBFE calculation method could save drug designers an enormous amount of time and resources by directing more attention to candidate molecules worthy of further experimental testing.[3,4] Various scoring functions and molecular dynamics (MD) simulations are the mainstream techniques to calculate RBFE. However, considering that LO requires high-precision estimation of RBFE for a small number of structurally related compounds, MD-based RBFE calculation methods should be a better choice than simple scoring functions.[5,6]

The basic principle of MD-based RBFE calculations is that by simulating the protein−ligand binding process, one can calculate binding thermodynamic properties, such as RBFE, within the rigorous formalism of statistical mechanics. Due to the incorporation of microscopic and physical detail, such as solvent-induced interactions and conformational dynamics,

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Table 1. Overall Relative and Absolute Binding Free Energy Calculation Results

metrics		system							
		BACE[f]	CDK2	JNK1	MCL1[f]	p38	PTP1B	thrombin	TYK2
crystal structure[a]		4DJW	1H1Q	2GMX	4HW3	3FLY	2QBS	2ZFF	4GIH
no. of compounds[b]		33	14	16	30	33	20	5	15
binding affinity range (kcal/mol)[c]		3.50	3.42	3.39	4.18	3.78	4.08	0.84	4.28
$\Delta\Delta G$[d]	MUE (kcal/mol)	0.84	1.11	1.15	1.21	1.32	1.52	0.29	1.55
	Pearson R	0.55	0.46	0.08	0.14	0.02	0.17	0.88	0.43
ΔG[e]	MUE (kcal/mol)	0.85	0.84	0.71	1.37	1.33	1.24	0.22	1.08
	Pearson R	0.71	0.46	0.08	0.55	0.02	0.17	0.88	0.43

[a]The crystal structure used as the reference of binding mode for each protein−ligand data set. [b]The number of ligands per protein−ligand data set. [c]The range of the experimental binding affinities per protein−ligand data set. [d]Three evaluation metrics—MUE and Pearson R—from the perspective of $\Delta\Delta G$. [e]Three evaluation metrics—MUE and Pearson R—from the perspective of ΔG. [f]The perturbations of the two systems are connected by both computational and experimental values, leading to the incomparability with analogous systems in other studies.[5,22,47]

MD-based RBFE calculations could be accurate enough to guide LO in principle.[4,7] However, achieving a sufficiently convergent RBFE calculation through brute-force MD simulations is quite challenging because of the high-dimensional potential energy surface that needs to be sampled.[8] In the face of this dilemma, alchemical free energy (AFE) methods, based on the notion of order parameters, stand out for their remarkable performance and extensive track record of successful applications over the past few decades.[4,9,10]

Free energy perturbation (FEP), one of the AFE methods, introduces a series of virtual intermediate states between the two end points through the coupling variable λ, bridging the overlap between adjacent phase spaces.[11,12] Thermodynamic integration (TI), another AFE method, differs from FEP in how to bridge the phase spaces when calculating RBFE. While FEP calculates RBFE by summing up the changes in free energies between a series of adjacent states, TI calculates RBFE by integrating the derivative of the thermally averaged potential energy of a state λ with respect to λ.[13,14] Nevertheless, the balance between the accuracy and the cost of these computational methods has not been optimal, and it hinders a wider adoption of FEP and TI in practical applications due to their enormous consumption of computational resources, despite their supposedly highly accurate modeling capacity.[5,15,16] To overcome these challenges, the λ-dynamics (LD) approach was first proposed in 1996, where the coupling variable λ is considered to be a dynamic variable with a fictitious mass.[17] In this way, a single MD simulation is enough to automatically sample almost all values of λ, while FEP and TI generally require running 12, 15, or more MD simulations for each preset λ intermediate. In addition, LD can include a series of congeneric ligands in a single MD simulation by assigning different λ variables to different substituents.[18] Furthermore, multisite λ-dynamics (MSLD) has been proposed to handle a series of congeneric ligands with multiple substitution sites.[19] Hitherto, 迄今为止 MSLD has become a powerful emerging method and an attractive alternative to performing a rational LO.

Hayes et al. constructed a basic λ-dynamics engine (BLaDE), enabling MSLD simulations on graphics processing units (GPUs) and resulting in 5 to 8 times faster than CHARMM-based MSLD simulations.[20] For this reason, there are two MD engines for MSLD: CHARMM and BLaDE standalone.[21] Nevertheless, an open-source MSLD procedure based on the BLaDE engine has not yet been established yet. In addition, the laborious procedure of multiple topology model (MTM) constructions for ligands and system setup has impeded the continual development of MSLD in both its practical implementation and its theoretical expansion. Regarding this issue, Raman et al. reported a workflow to set up, run, and analyze MSLD simulations using BIOVIA Discovery Studio and Pipeline Pilot (referred to as the DS workflow henceforth).[22,23] This workflow, however, heavily relies on commercial software. Although some works have reported how to use alternative software tools to build MTM and execute MSLD simulations automatically,[24,25] the workflow based on these components requires lots of case-by-case tinkering as it involves various dynamic engines and scripts. Besides, the software packages commonly used for RBFE calculations, such as CHARMM,[21] NAMD,[26] AMBER,[27] and GROMACS,[28] pose challenges for both novices (including experimenters) and developers. First, some of these software packages lack concise tutorials, making it tough for novices or experimenters to get started. Second, the limited modularity and openness of certain packages make it challenging to test and combine different methods or develop custom features.

Herein, we present a comprehensive, open-source, and automated workflow for MSLD based on BLaDE standalone. It covers protein and ligand preparation, MSLD simulations, and postprocessing. In particular, we introduce a ligand Internal and Cartesian coordinate reconstruction-based alignment algorithm (LIC-align) in the ligand preparation stage and optimize the maximum substructure (MCS) search algorithm in the MTM creation stage to improve the accuracy of the MSLD input generation. We also provide the scripts to connect various components more robustly, making all results reproducible and requiring minimal user's intervention. The proposed workflow is specifically set up in a modular fashion to facilitate straightforward combinations and extensions of various techniques. Finally, the workflow is fully accessible and novice-friendly, with many automated procedures. To validate the reliability and efficiency of our workflow, we use it to calculate the RBFE for 166 ligands across 8 proteins. We also analyze and provide empirical computing strategies for a broad range of application scenarios, such as multisite perturbation, large perturbation, scaffold hopping, and multiattachment perturbation. Novices or experimenters can achieve trustworthy calculation results using the default protocols and parameters in our workflow. Compared to the commercial DS workflow, our workflow adopts more reasonable and appropriate methods of processing in certain cases.

■ METHODS

Data Sets. The validation data set was established based on a classical FEP benchmark reported by Wang et al. in 2015.[5] It

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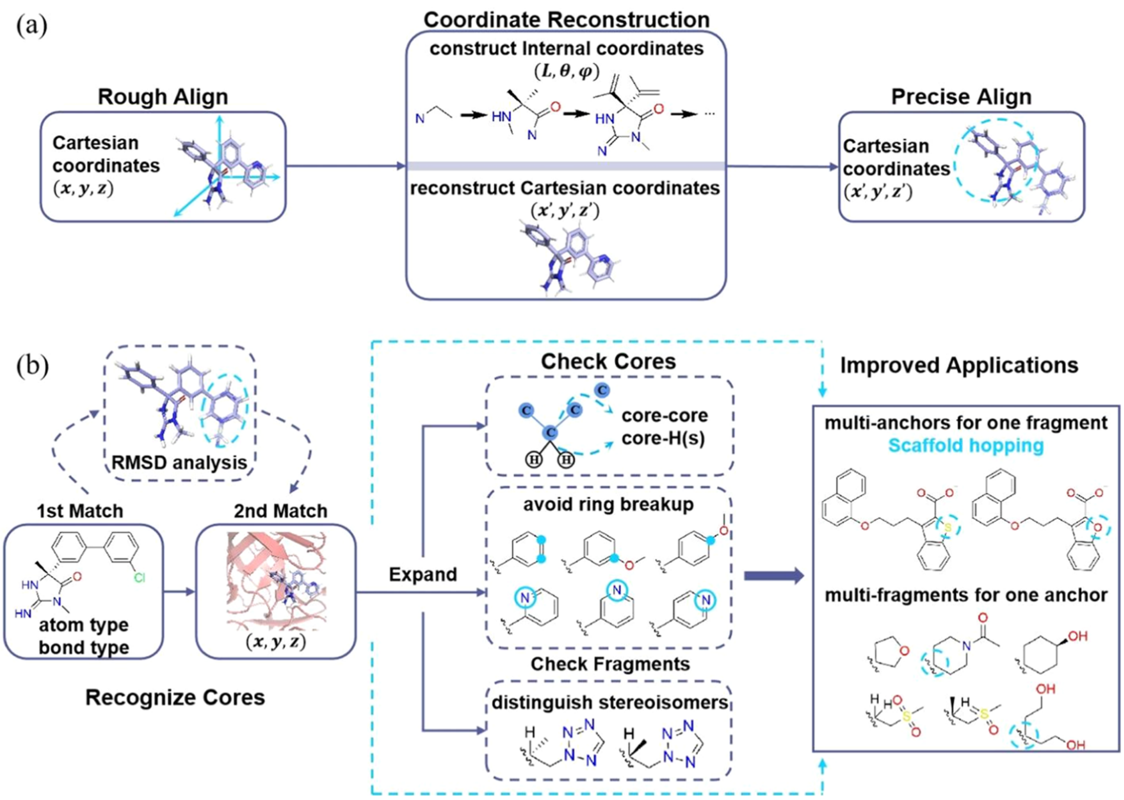

Figure 1. (a) LIC-align algorithm. The ligands are roughly aligned using the Cartesian coordinates in the first step and are then precisely aligned using the reconstructed Cartesian coordinates in the subsequent two steps. Depending on demand, users may choose to enable or disable the following two steps to achieve precise alignment and rough alignment, respectively. (b) The MCS search algorithm. It is possible to establish the ideal perturbation mode and investigate other modes, such as single-site multiattachment perturbation (see JNK1−1 subset, JNK1−2 subset, etc.) and scaffold hopping (see MCL1−1 subset). Scaffold hopping involves the modification or replacement of a molecule scaffold (usually a ring), which refers to the perturbation of several atoms attached to multianchors. Multiattachments (multifragments for one anchor) refer to the situation where an anchor atom can be attached to multiple heavy atoms or hydrogen atoms.

contains ligand binding data for eight targets, including BACE, CDK2, JNK1, MCL1, p38, PTP1B, thrombin, and TYK2, as shown in Table 1. Due to the incompatibility between the current coordinate compressing step of BLaDE and the required processing of lone pair electrons when employing a small molecule force field, such as CHARMM general force field version 2.5 (CGenFF v2.5),[29] the ligands containing lone pair electrons in their perturbation groups were excluded. As a result, 8 protein−ligand systems with 166 ligands in total were included in the validation data set. To carry out MSLD calculations efficiently, the data set was divided into 29 subsets. The grouping strategy was elaborated in the Supporting Information (SI) and the following discussion section. All ligands and how they are grouped are summarized in Figures S1−S8. It should be noted that due to the computational complexity introduced by lone pairs, the ligands for the BACE and MCL1 systems cannot be fully connected by calculations when grouped. To be specific, within these two systems, certain ligands have lone pairs while others do not, so connecting ligands with and without lone pairs would necessitate introducing lone pairs in the perturbation region, which is currently unfeasible. Therefore, the disconnected parts need to be linked by experimental values (specifically, 4 values). The experimental binding affinity of each ligand was taken from the same references used by Wang et al.[5,30−37]

Workflow for MSLD. *Protein Preparation.* The protein structures were downloaded from the Protein Data Bank (PDB),[38] as shown in Table 1. The proteins were optimized using the Protein Preparation module in Schrödinger (Schrödinger Inc., NY), in which the missing side chains and loops were rebuilt, the pK_a values of the ionized residues in proteins were set at pH = 7.4, and the structures were minimized with the OPLS2005 force field. Then, CHARMM was used to solvate the complex (protein−ligand) in a cubic box with the periodic boundary edge set to 10.0 Å, where the selected explicit water model was TIP3P and the counterions were 0.15 mM NaCl.[21,39] It should be noted that 0.15 mM KCl was used as the counterions in the thrombin systems because sodium ion binding to specific sites would induce allosteric changes in thrombin, whereas potassium ions could maintain the stability of the system.[40,41] The force fields for the proteins and ligands were CHARMM36m and CGenFF, respectively.[29,42] It is important to note that Schrödinger is not an open source, but the community offers numerous alternative open-source tools for protein preparation, such as GROMACS[28] and CHARMM-GUI,[59] which can be selected based on individual resources.

Ligand Preparation. The ligand structures were obtained from the SI of the work published by Wang et al.[5] With reference to the structures of the proteins, the ligands were first positioned at the binding sites. Then, the ligands per subset were rigidly aligned, in accordance with their MCS(es). The perturbation patterns that show how their MCS and perturbation groups were defined are summarized in Figures S1−S8. Both steps of the spatial alignment were accomplished by using the LIC-align algorithm. The force field parameters and topologies of each ligand were generated using CGenFF v2.5.[29]

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LIC-align Algorithm. The core algorithmic logic of the LIC-align algorithm is depicted in Figure 1a. First, the MCS of the ligands is searched using the rdFMCS module of the RDKit package.[43] When searching the MCS, the "completeRingsOnly" option is enabled to avoid breaking rings. Then, the ligands are aligned according to the coordinates of their MCS by invoking the Kabsch rotation.[44] Suppose the MCS contains more than three atoms, in that case, the algorithm would try to precisely align the ligands to ensure that the MCS in each ligand contains the same coordinates while keeping their internal structure unchanged. Such a "precise alignment" is done by the following steps: (1) First, three connected atoms are selected from the MCS as the "base atoms" to construct a Z-matrix coordinate system and (2) then, using the selected atoms, the Cartesian coordinates of the target ligand (which contains the target MCS structure) and the aligned ligand are converted to Z-matrix coordinates. When this conversion is performed, the atom in the MCS is placed at the top of the Z-matrix coordinate array and the order of atoms in the MCS of the two ligands is kept the same. Besides, during the conversion, we always construct the bond length component of the Z-matrix coordinate of the newly added atom based on an atom that is connected with it. Such measures ensure that the internal structure of the ligands remains unchanged during the conversion. After the conversion, we directly set the Z-matrix coordinates of the MCS in the aligned ligand to the ones in the target ligand and convert the Z-matrix coordinates of the aligned ligand back to Cartesian coordinates. By doing so, even if rough alignment or most alignment software cannot achieve perfect alignment of the MCS, we can slightly adjust the structures to ensure strict consistency of the MCS while preserving the correct relative position of the MCS and substituents. However, since the restoration of the Cartesian coordinates would change the absolute Cartesian position of the ligand, we invoke the Kabsch rotation of the aligned ligand again using the coordinates of its MCS.

Multiple Topology Model. Similar to FEP and TI calculations, the topology of the MCS (also referred to as the core) per subset was single, while the topology of the perturbation groups (also referred to as the fragment) was dual or multiple. The anchor atom that belonged to the core was connected to the topologies of the core and the fragment. To automatically create such a complicated MTM, the MCS and perturbation groups must be accurately recognized by our modified MCS search algorithm first. After that, the atomic charges of each MCS were recalculated by the charge renormalization algorithm described in ref24, resulting in an integer net charge for each end point representing a physical ligand. The charges of each ligand summing to an integer net charge must be strictly imposed in MSLD.[45] Once the charge renormalization was completed, the intermolecular force field parameters were assigned using Lennard-Jones force field parameters,[46] with the original intramolecular force field parameters retained. Finally, the MTM-related files were generated, which were system-specific and could be readily connected to the subsequent stage.

Modified MCS Search Algorithm. This paragraph covers the underlying logic of our proposed algorithm that is partly adapted from the original MCS search algorithm[24] and it applies more appropriate strategies and functions during MSLD calculations. From Figure 1b, we can see the motivation for the algorithmic design used to achieve these functionalities. The first match identifies core atoms based on identical force field atom types and lists of bonded neighbor atom types. Then, several redundancy and error checks are conducted to confirm that (1) each molecule has the same number of core atoms, (2) each core atom has one and only one corresponding core atom in other molecules, and (3) each core atom is not matched repeatedly. After that, an analysis verifies that all matched atoms have coordinate root-mean-square deviation (RMSD) values within 0.8 Å. The second match identifies core atoms among the remaining atoms according to identical force field atom types and spatial distances (coordinate RMSD values within 0.8 Å). Strict alignment of the MCS between congeneric ligands is required for accurate identification of the second match, which could be accomplished using the previously mentioned LIC-align algorithm. The redundancy and error checks mentioned in the first match are then repeated. Besides, some other checks are added to ensure that (1) core atoms identified in this step are linked to those recognized in the first match and (2) matched atoms (excluding the anchor atom) connect to the same number of hydrogen atoms. Atoms that fail to pass the checks would be removed from the MCS. Also, considering that ring breakage and reformation are detrimental to the statistical mechanical rigor of alchemical calculations,[47] conjugated cycles with multiple substitution sites or different heteroatoms are deleted from the MCS and moved into perturbation groups. This is accomplished by recognizing the conjugated rings and their internal and external connections in the MCS. Following MCS recognition, the leftover atoms are classified as fragments. Taking the flexibility of single bonds into account, all hydrogen atoms attached to the anchor atom are classified as fragments or the constraint of such single bonds would be overly rigid. Apart from that, the anchor atom could be attached to heavy atom(s) and hydrogen atom(s) singly or multiply, leading to the accomplishment of "multianchors for one fragment" and "multifragments for one anchor." Subsequently, fragment redundancy is evaluated to ensure that the same fragment would not be sampled repeatedly. Such a redundancy check of fragments is based on not only force field atom types but also coordinates to ensure that stereoisomers are not regarded as identical. Finally, the output is printed as a prepared text file for the next stage. Overall, these underlying algorithms primarily resulted in two expanded uses for the MCS search algorithm: (1) scaffold hopping and (2) exploration of more perturbation patterns because the anchor atom could attach to multiple heavy atoms or hydrogen atoms.

Simulation System Setup. By combining the MTM of the ligands per subset with the corresponding protein with periodic boundary conditions, the simulation system was built up using CHARMM.[21,39] Two systems were generated to simulate the complex in solution and the ligands in solution independently for RBFE calculation. The only difference between the two systems was whether the protein was included. The same processing included (1) reading in preprepared periodic boundary conditions, the MCS, and all perturbation groups, (2) patching all of them into the protein structure file, and (3) deleting internal energy terms that span two substituents in the same site. To eliminate potential local steric clashes in the system, the steepest descent minimization for a maximum of 1000 steps and the conjugated gradient minimization for a maximum of 200 steps were performed continuously until the gradient root-mean-square (GRMS) was under 3 and 0.1, respectively. Subsequently, the coordinates, topologies, and structures of the simulation system were obtained.

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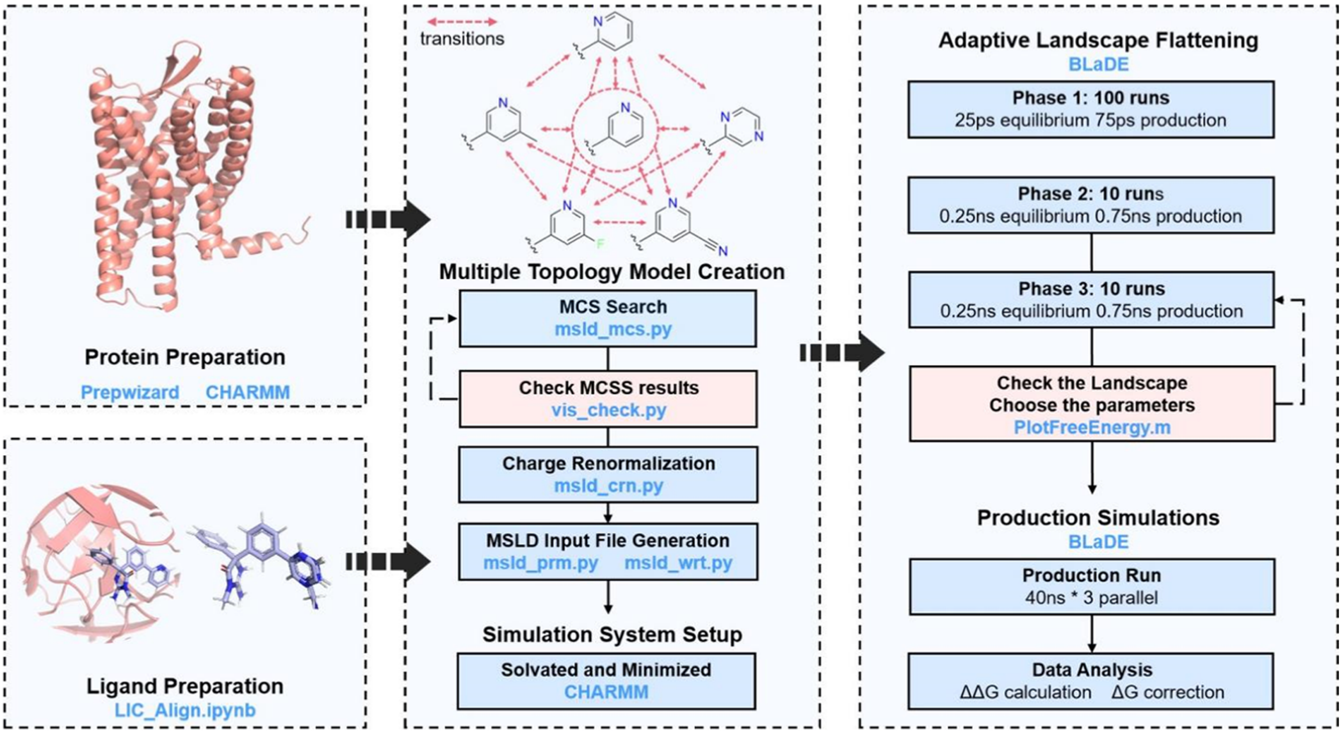

Figure 2. Comprehensive, open-source, and automated workflow for MSLD simulations based on BLaDE standalone. Following protein and ligand preparation, this workflow first creates a multiple topology model for a subset of ligands, then sets up the simulation system, and finally performs ALF simulations and production simulations to calculate $\Delta\Delta G$ and ΔG. It is important to note that the *prepwizard* module in Schrödinger is not freely available, but the community offers a wide range of alternative open-source protein preparation tools for researchers.

MSLD Simulations. Once set up, all simulations were performed using the BLaDE standalone, a GPU-based dynamics engine that is 5 to 8 times faster than the domain decomposition (DOMDEC) in CHARMM.[20] To efficiently acquire the MSLD calculation results, the MSLD simulations were performed in two steps: ALF runs and production runs.[25] The ALF simulations were performed in three phases: (1) 100 iterations of 100 ps short MSLD simulations, (2) 10 iterations of 1 ns MSLD simulations, and (3) 5 iterations of 2 ns long MSLD simulations.[22] With respect to the production simulations, 3 parallel simulations using different random seeds were performed, with each having 5 ns of equilibrium and 35 ns of production. For a subset of ligands, two independent MSLD simulations were performed: one for the complexes in the solvent and the other for the ligands in the solvent.

All simulations were performed in the NPT ensemble with a time step of 2 fs. The temperature was kept at 298.15 K by Langevin dynamics with a friction coefficient of 5 ps^{-1}.[48] The pressure was controlled at 1 bar using a Monte Carlo barostat.[49,50] The holonomic constraint variable fnex was set to 5.5 to ensure sufficient sampling of end points, and the soft-core potentials were used to prevent end point singularities. During the equilibration and production, a switching distance was set to 10 Å, and the cutoff distance of nonbonded reactions was set to 12 Å. The particle mesh Ewald (PME) algorithm was used to treat the long-range electrostatic interactions,[51−53] the fast SHAKE algorithm was used to restrain hydrogen bond lengths in ligands and proteins, and the SETTLE algorithm was used to restrain bond lengths in water.[54,58] It is worth mentioning that the parameters in the bias potential were optimized iteratively during the ALF simulations, and the best parameters obtained were used in the production simulation.

Data Analysis. $\Delta\Delta G$ (the calculated relative binding free energy) between one ligand and another was obtained by

$$\Delta\Delta G_{\text{bind}}(\lambda_{s,i} \Rightarrow \lambda_{s,j}) = -k_B T \ln \frac{P(\lambda_{s,j} > \lambda_c)}{P(\lambda_{s,i} > \lambda_c)} - \Delta U_{\text{bias}} \quad (1)$$

where $\lambda_{s,i}$ corresponds to substituent i in site s, $\lambda_{s,j}$ corresponds to substituent j in site s, k_B is the Boltzmann constant, T is the temperature of the system, $P(\lambda_{s,j} > \lambda_c)$ is the probability of the occurrence of ligand j defined by the frequency when $\lambda_{s,j} > \lambda_c$, $P(\lambda_{s,i} > \lambda_c)$ is the probability of the occurrence of ligand i defined by the frequency when $\lambda_{s,i} > \lambda_c$, and ΔU_{bias} is the bias potential. Equation 1 is the classical definition of binding free energy in statistical thermodynamics.

ΔG (the calculated absolute binding free energy) of one certain ligand was obtained by

$$\Delta G_{\text{calc}} = \Delta\Delta G_{\text{calc}} - \left(\frac{\sum \Delta\Delta G_{\text{calc}} - \sum \Delta G_{\text{exp}}}{n} \right) \quad (2)$$

where ΔG_{calc} is the calculated absolute binding free energy (ABFE) and ΔG_{exp} is the experimental ABFE. Equation 2 refers to using all experimental values of the ligands in the whole protein−ligand system as a reference, rather than fixing a specific ligand as a reference, as was done previously.[5,22] This approach helps to eliminate the propagation of errors that may exist in any one experimental value.

Evaluation and Analysis. Finally, an evaluation of the accuracy and efficiency of the calculations obtained by this automated BLaDE-based workflow was provided. The adopted evaluation metrics were the mean unsigned error (MUE) and Pearson correlation coefficient (Pearson R). The analysis of

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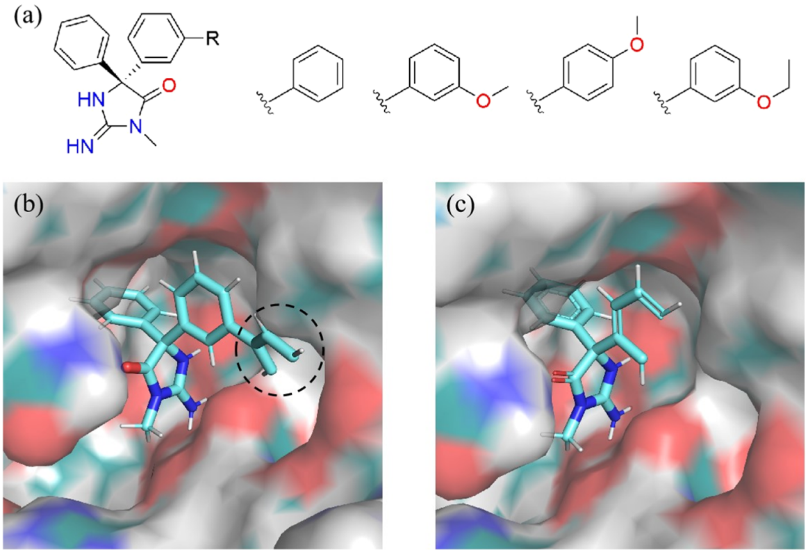

Figure 3. Core atoms (without anchor atoms) recognized by the MCS search algorithm. (a) The BACE-3 system consists of 4 ligands, in which the substituted phenyl is defined as the perturbation group. (b) Core atoms recognized by the MCS search algorithm before optimization. (c) Core atoms recognized by the optimized MCS search algorithm.

efficiency started from the required simulation time scale to the wall time needed to prepare the input and perform all simulations.

■ RESULTS AND DISCUSSION

The main obstacles and concerns related to MSLD are, as already mentioned, its intricate preparation and calculation procedure and its accuracy and efficiency, especially compared to FEP. Therefore, we first give an overview of our well-established workflow, which is easy to use by following our guidelines in GitHub (see "data and code availability") and especially illustrates how the LIC-align algorithm and the MCS search algorithm are beneficial to the MTM creation. After that, we demonstrate the reliability of our comprehensive, open-source, and automated workflow by comparing the calculated $\Delta\Delta G$ and ΔG with the experimental binding affinities from both overall and group perspectives. Through a few case studies, we analyze the variables that influence the sampling and accuracy of MSLD calculations and propose some empirical calculation strategies. Since these empirical strategies have been incorporated into our workflow, an extensive analysis was conducted to find out the difference between our workflow and the DS workflow when handling intricate ligand scenarios. We also compare the accuracy and efficiency of MSLD and FEP through a detailed case study of thrombin.

Workflow for MSLD. The workflow is comprehensive, open-source, and automated, covering protein and ligand preparation, MTM creation, simulation system setup, ALF, and production simulations. It not only incorporates all available open-source components but also includes the LIC-align algorithm for ligand preparation and the optimized MCS search algorithm for MTM creation as well as the scripts to connect components flawlessly and robustly. The modular establishment of the workflow facilitates the combination and advancement of diverse techniques. Moreover, the workflow is readily available to the community and is constructed to be straightforward to use for individuals who are inexperienced, which considerably simplifies MSLD calculations. Figure 2 illustrates the stages in the workflow and the main algorithms, scripts, and software used at each stage. The "Methods" Section provides an explanation of each stage in this workflow, including the important parameters and calculation methods. To facilitate the utilization of our workflow by researchers, comprehensive guidelines are provided in a step-by-step format in the Github repository. Additionally, sample scripts have been included in the SI to demonstrate the automated and convenient usage of our workflow.

The combination of the LIC-align algorithm and the optimized MCS search algorithm allows for more accurate recognition of the MCS and various perturbation patterns. To be specific, as it was difficult for the common alignment algorithms to accomplish strict alignment of the MCS, the original MCS search algorithm adopted spatial matching as a last resort. The consequence is a rather imprecise recognition of the maximal substructure based on atom types and the surrounding chemical environment, as atoms possessing comparable chemical environments tend to end up with high ambiguity. However, as the LIC-align algorithm achieves a strict alignment of the MCS by mutually converting Cartesian coordinates and internal coordinates, we increase the priority of the spatial matching to achieve a precise identification of the MCS. In addition, the functional modules of multisite perturbations, scaffold hopping, and multiattachment perturbation groups for one site have been enhanced to enable users to investigate more diverse perturbation patterns. Because of the implementation of these two algorithms, manual editing and adjustment of certain procedures are no longer necessary, highlighting the convenience of our workflow.

Our workflow exhibits superior performance when handling challenging ligand sets. The issue of chiral isomers has long been an uphill matter in the MTM creation stage. The original MCS search algorithm lacks the ability to differentiate between chiral isomers as it does not consider spatial coordinates while removing the redundancy of fragments. Furthermore, the fundamental reason for not considering spatial coordinates is that different fragments have the probability of being incorrectly deemed redundant without a precise alignment originally. However, our workflow benefits from utilization of the LIC-align algorithm, which allows for the differentiation between

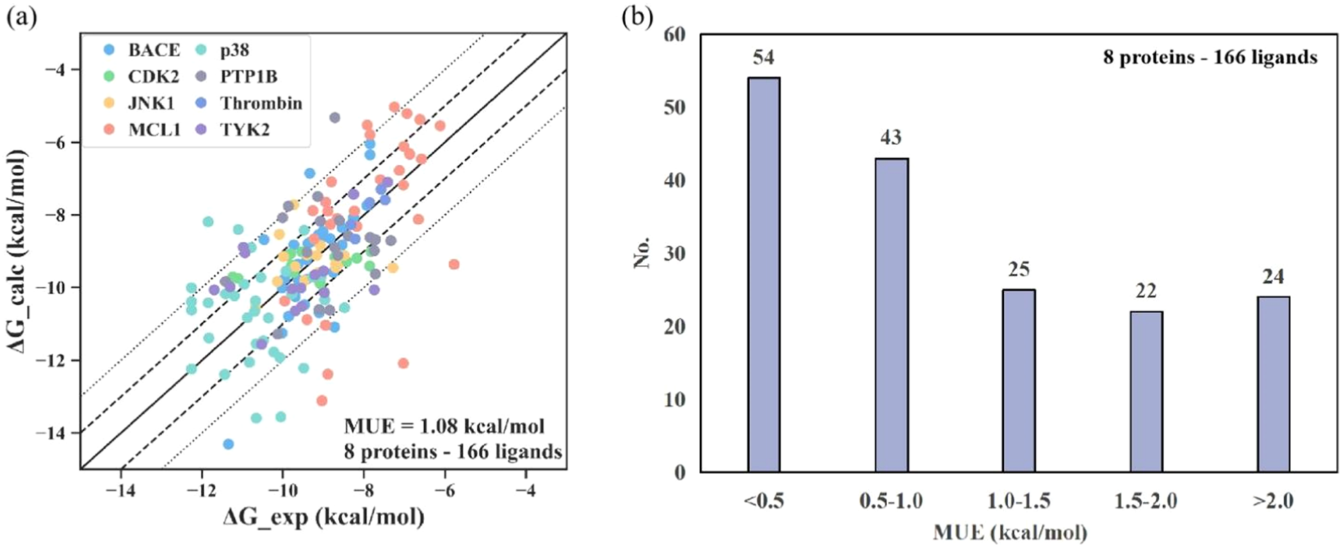

Figure 4. Overall absolute binding free energy calculation results. (a) Correlation between the calculated and experimental ΔG of all protein–ligand complexes, with the overall MUE on the bottom right (the perturbations of 6 targets are fully connected via the calculations, while the perturbations of BACE and MCL1 are connected via both the computational and experimental data). Each protein–ligand system corresponds to a type of dot with different colors. Dots on the solid line indicate that the calculated values are completely correlated with the experimental values; dots within the area of the dashed line indicate that the errors are less than 1.0 kcal/mol, and dots within the area of the dotted line indicate that the errors are less than 2.0 kcal/mol. (b) Distribution of the MUEs for all ligands. The number of ligands belonging to this class is labeled on the top of each bar.

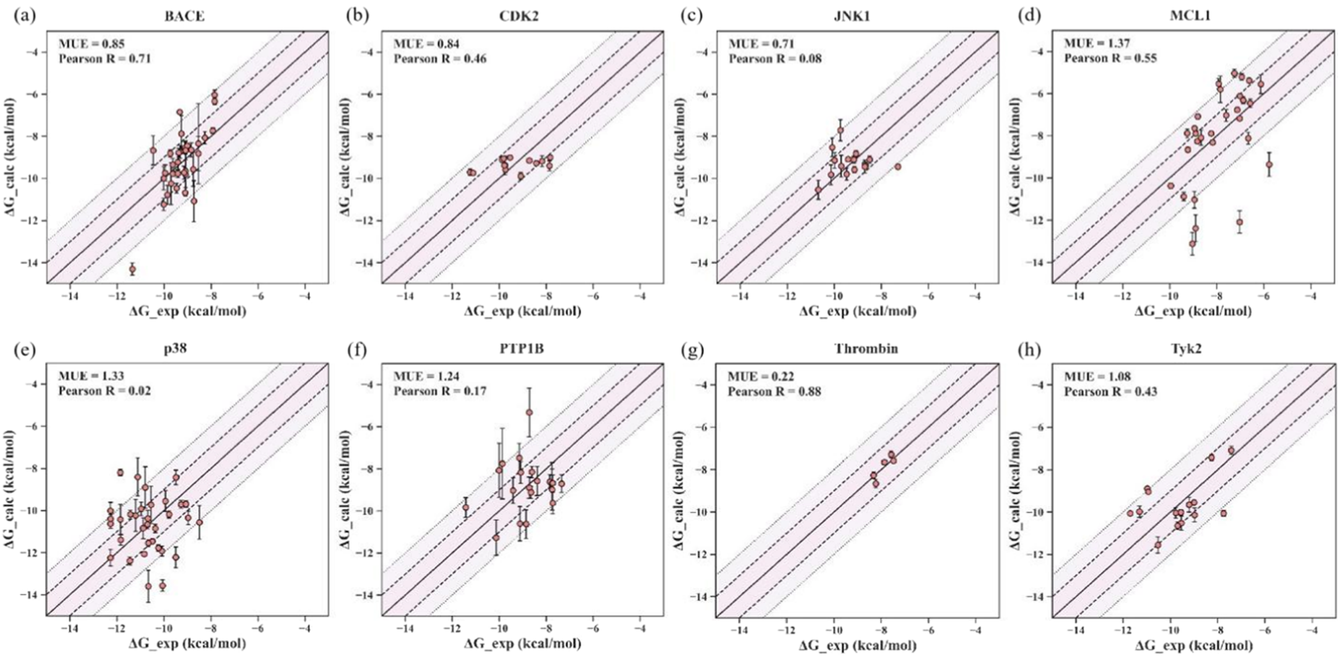

Figure 5. Correlation between the calculated and experimental ΔG for 8 protein–ligand systems (a–h), with the name of systems above the panels, the MUE and the Pearson R of each system are on the top left. Dots on the solid line indicate that the calculated values are completely correlated with the experimental values; dots within the area of the dashed line indicate that the error is less than 1.0 kcal/mol, and dots within the area of the dotted line indicate that the error is less than 2.0 kcal/mol. Panels (a–h) correspond to the BACE, CDK2, JNK1, MCL1, p38, PTP1B, thrombin, and Tyk2 systems, respectively.

chiral isomers through spatial position confirmation. Furthermore, as proven below, our workflow endeavors to preserve the integrity of rings. As mentioned before, ring breakage and reformation appear to violate the statistical mechanical rigor of alchemical simulations.[47] This is one of the crucial factors taken into consideration by our optimized MSC algorithm, which makes sure that conjugated cycles with multiple substitution sites or different heteroatoms are not cut off when recognizing the core atoms. The older algorithm, on the other hand, breaks the ring in order to fit as many atoms as possible into the core. As an illustration, the BACE-2 system involves four ligands, where the substituted phenyl group is regarded as the perturbation group. The original MCS search algorithm broke the benzene ring while identifying the core atoms and incorporated the ortho-position and one of the meta-position atoms into the core, as illustrated in Figure 3b. On the other hand, the recognition of

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core atoms follows the ideal pattern after the algorithm has been improved, as shown in Figure 3c. Keep in mind that anchor atoms are not visible in Figure 3c, implying that the identified core atoms comprise a full ring. For the reasons mentioned above, our LIC-align algorithm and the optimized MCS search algorithm provide favorable alternatives to prepare the ideal MSLD input files more accurately.

Overall Performance of MSLD Calculations. The $\Delta\Delta G$ and ΔG of 166 ligands across 8 protein−ligand data sets were calculated using the proposed workflow. The left panel of Figure 4 shows that there is a substantial correlation between the calculated ΔG and experimental binding affinities with an MUE of 1.04 kcal/mol. Furthermore, the right panel of Figure 4 presents that the ligands with a deviation of less than 1.0 kcal/mol from the experimental values account for more than 57.1%. This can also be seen from the left panel of Figure 4 since the errors of almost all ligands are within 2.0 kcal/mol (dotted lines) and the errors of most ligands are within 1.0 kcal/mol (dashed lines). It should be mentioned that the ligands for BACE and MCL1 are not fully computationally connected, and some tough parts are connected by experimental data. While the overall accuracy cannot be directly compared with that reported by other studies that have also examined these systems,[5,22,47] the overall results still reflect the accuracy and efficiency of our workflow.

These results come from 8 data sets, and each data set represents a protein system. As shown in Table 1, in terms of $\Delta\Delta G$, the MUE varies from 0.29 to 1.52 kcal/mol, and the lowest Pearson R is 0.02. The MUE for ΔG ranges from 0.22 to 1.37 kcal/mol, with 0.02 being the lowest Pearson R.

Of interest here is the increase in Pearson R after $\Delta\Delta G$ was converted to ΔG for the BACE and MCL1 system. This could be attributed to the fact that the ligands for BACE and MCL1 are not fully connected by the computational data. Furthermore, when experimental values are applied to connect subsets, larger or more difficult perturbations may be introduced, leading to overestimated results. Figure S9 shows that when the subsets are connected by the experimental values rather than the computational values from each other, both the MUE and the Pearson R increase. However, as mentioned in the previous "Methods" Section, due to the computational limitations imposed by lone pairs, we are unable to fully connect the ligands for BACE and MCL1. Therefore, readers are recommended to refer to the calculation results of each subset rather than those of the entire BACE or MCL1 system.

The correlation between the calculated and experimental ΔG for every system is set out in Figure 5. The estimated standard deviations of $\Delta\Delta G$ and ΔG were obtained by bootstrapping. As can be seen from panels (a−f), there are a few data points with standard deviations exceeding 1.0 kcal/mol, demonstrating the superior reproducibility of our workflow. Probable rationales of the relatively high standard deviation are that (1) the perturbation group corresponding to the data point changes significantly in structure from other perturbation groups in the same subset (accounts for 4 out of 5 ligands with standard deviation exceeding 1.0 kcal/mol) and (2) the perturbation groups included in the corresponding subset could be classified as large perturbation groups (accounts for 1 out of 5 ligands with standard deviation exceeding 1.0 kcal/mol). Both of these issues might result in slow convergence and inadequate sampling of the MSLD simulations, which would further lead to considerable fluctuation in the 3 parallel samplings. Concerning this, it is worth the efforts to (1) group more precisely according to the size of the perturbation group, (2) divide the subset with multiple large perturbation groups into 2−3 smaller subsets, and (3) increase the number of parallel simulations, such as 5 or 8 replicas, to guarantee efficient sampling in as many replicas as possible. This finding confirms the practicality of the grouping strategy mentioned in the SI given that grouping based on the similarity of ligand structures could ensure the size of each perturbation group is as small as possible. Likewise, it reinforces the rationality of the strategy of grouping upon the size of perturbation groups, in line with the statements made in prior studies.[55] The implementation of comparable perturbation groups within a given subset could potentially mitigate significant changes from small or medium perturbations to large ones, further preventing high potential energy barriers that separate phase space.

What emerges from the results reported above is that our comprehensive, open-source, and automated workflow is reliable and efficient enough to perform MSLD calculations. First, the overall mean unsigned error (MUE) of 1.04 kcal/mol and the MUEs for each target or subset obtained through our workflow are reasonably accurate. However, it should be noted that the validation data sets used in some similar evaluation studies are different,[5,22,57] as are the definitions of perturbation groups and data set division (especially BACE and MCL1 are not fully connected in our study). It is therefore difficult to either compare these results or determine which workflow is definitely better in terms of accuracy. Our results may only demonstrate that our workflow is capable of achieving highly accurate results. Additionaly, the $\Delta\Delta G$ of a subset with less than 10 congeneric ligands could be calculated in as short as 150 ns through our workflow, the same as that through the DS workflow.[22] Interestingly, 3 replicas of 40 ns are used in our workflow, whereas 6 replicas of 20 ns are used in the DS workflow. This may have an impact on the standard deviation between parallel results, as the standard deviations for these two workflows are 0.47 and 0.29 kcal/mol, respectively. The finding is consistent with what was mentioned in the last paragraph. In summary, the workflow reported here has significant potential for obtaining highly precise $\Delta\Delta G$ and ΔG to guide LO.

Performance of MSLD Calculations on Various Subsets. The 8 protein−ligand systems were further divided into smaller subsets in accordance with the criteria outlined in the SI. The fundamental information and the MUE of each subset are summarized in Table 2. It can be seen from the data that our workflow performs outstandingly for a variety of systems and subsets. Moreover, different scenarios, such as single-site or multisite substitution, large substituent perturbation or small substituent perturbation, and multiattachment perturbation or scaffold perturbation, are covered in these subgroups. Below, these different scenarios are discussed, which could help us to identify the crucial parameters that affect MSLD calculations and thus propose appropriate empirical strategies to strengthen the robustness of MSLD calculations.

Single Site or Multisite: CDK2 Case Study. The multisite perturbation system considers all possible combinations of substituents at different sites for an MSLD simulation. For example, JNK1−2 is a two-site system, with site 1 having 5 substituents and site 2 having 4 substituents. After the MSLD simulations, 20 results would be obtained, since there are 20 possible combinations of substituents. Similarly, 18 results were acquired for the MCL1−7 subset, which is a three-site system.

The CDK2−1 system consists of 6 ligands that were modeled using the single-site and three-site methods separately, as shown

Table 2. ΔΔ*G* Calculation Results for the Subsets per System

subset[a]	N site[b]	M subs[c]	N exp[d]	R size[e]	range (kcal/-mol)[f]	MUE (kcal/-mol)[g]
BACE-1	1	9	9	M	1.63	0.60
BACE-2	1	5	5	L	1.16	0.95
BACE-3	1	4	4	L	1.78	0.86
BACE-4	1	7	7	L	1.53	0.81
BACE-5	1	8	8	L	1.93	1.04
CDK2−1	1	6	6	L	1.92	0.48
CDK2−2	1	9	9	S	3.07	1.41
JNK1−1	1	9	9	L	2.70	0.64
JNK1−2	2	5 × 4	7	S	1.98	0.67
MCL1−1	1	3	3	S	0.66	0.22
MCL1−2	1	3	3	L	0.82	0.61
MCL1−3	1	5	5	M	1.80	1.06
MCL1−4	1	5	5	M	1.01	0.28
MCL1−5	2	2 × 6	7	L	3.26	3.31
MCL1−6	2	2 × 2	3	S	0.73	1.01
MCL1−7	3	2 × 3 × 3	6	S	2.38	0.57
p38−1	1	8	8	L	2.40	1.48
p38−2	1	8	8	S	1.71	1.10
p38−3	1	5	5	M	1.77	1.07
p38−4	1	9	9	M	2.45	1.22
p38−5	1	7	7	L	2.03	0.96
PTP1B-1	1	6	6	S	1.74	0.45
PTP1B-2	1	6	6	M	1.41	0.92
PTP1B-3	1	5	5	M	2.7	0.51
PTP1B-4	1	6	6	L	1.42	0.93
thrombin-1	1	5	5	L	0.84	0.29
TYK2−1	1	4	4	S	0.76	0.41
TYK2−2	1	8	8	S	3.89	0.87
TYK2−3	1	5	5	M	3.56	1.91
					average[h]	1.04
					stdev.[i]	0.47

[a]The subset ID, with the protein−ligand system to which it belongs given before the dash. [b]The number of substitution sites. [c]The number of substituents per site. [d]The number of ligands with experimental binding affinities available. [e]The size of perturbation groups: small (S), medium (M), large (L). [f]The range of experimental binding affinities. [g]The MUE per subset. [h]The overall MUE of 166 ligands across 8 protein−ligand systems, with the unit kcal/mol. [i]The overall standard deviation of 166 ligands across 8 protein−ligand systems, with the unit kcal/mol.

in Figure 6. The former has a single substitution site and defines the whole substituted benzene ring as a perturbation group. The latter has three substitution sites, corresponding to the two meta-positions and one para-position of the benzene ring in the single-site method. The MSLD simulations showed that the sampling efficiency of the three-site method is significantly inferior to that of the single-site method. Additionally, the MUE calculated using the former method is 0.48 kcal/mol (see Table 2), while the one using the latter is unsuccessfully given because of the worse sampling efficiency. One possible explanation is that there are multiple combinations between the perturbation groups of different sites, though the latter has a small perturbation group, and therefore, there are additional potential energy barriers needed to overcome in the MSLD simulations. In addition to spanning potential energy barriers between substituents within a site, there are also potential energy barriers between sites that need to be spanned. Another probable explanation is that the anchor atom of the latter is defined on the

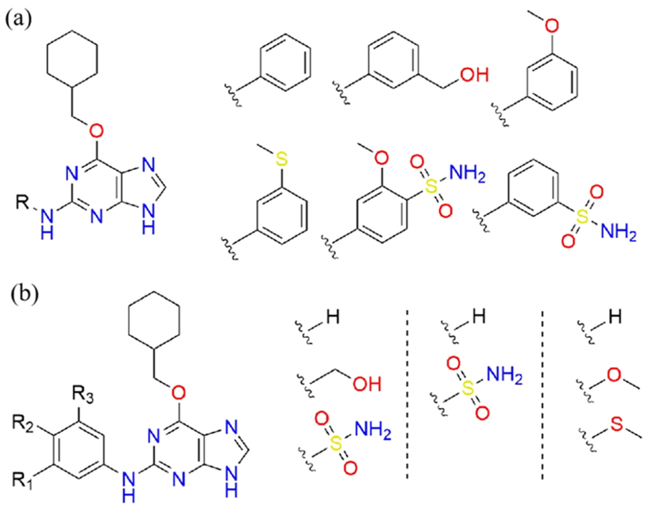

Figure 6. Two ways to define the perturbation groups in the CDK2−1 system: (a) one substitution site with relatively large perturbation groups and (b) three substitution sites with relatively small perturbation groups.

ring of the core, leading to the ring breaking and forming during the perturbation, which is highly undesirable based on an intuitive understanding of the MSLD algorithm. This point of view is covered in more detail in the following section. On top of that, even if MSLD calculations using the multisite definition can investigate as many ligands as feasible, not all of the combined ligands are of interest. Also, Raman et al. mentioned that additive estimates were interestingly more accurate than multisite estimates when dealing with large combinatorial libraries.[22] As a result, the single-site method is preferable for MSLD calculations. However, providing the option to carry out multisite calculations would still aid in the ongoing development and wider applicability of MSLD.

Categorize Ligands According to the Size of Perturbation Groups. Prior studies have shown that the grouping of subsets should take the size of the perturbation group into consideration.[55] Our own benchmark study on MSLD also supports this viewpoint. Take the PTP1B system as an instance, whose division of ligands and the definition of the perturbation groups are provided in Figure S6. During the MSLD simulations, initially, we divided the ligands into subsets on the basis of the chemical types of the perturbing groups, which triggered poor sampling, slow convergence, and (essentially) insensible output. However, the sampling was considerably improved, and meaningful results were obtained when the ligands were split by the size of the perturbation groups. This might be associated with the potential energy barriers required to be overcome in the transition between the perturbation groups in MSLD simulations, as the majority of the perturbation groups with comparable sizes necessitate the breaking and formation of only a few bonds. Overall, the size of the perturbation groups is an important factor, and categorizing ligands according to the size of their perturbation groups is a crucial tip for successful MSLD simulations.

How to Define the Anchor Atom: PTP1B Case Study. As already mentioned, a ligand can be split into two components: the core and fragment in the MTM stage. The atom that is attached to the fragments in the core is known as the anchor atom. By definition, anchor atoms belong to core atoms, yet they are perturbed along with fragment atoms in MSLD simulations.

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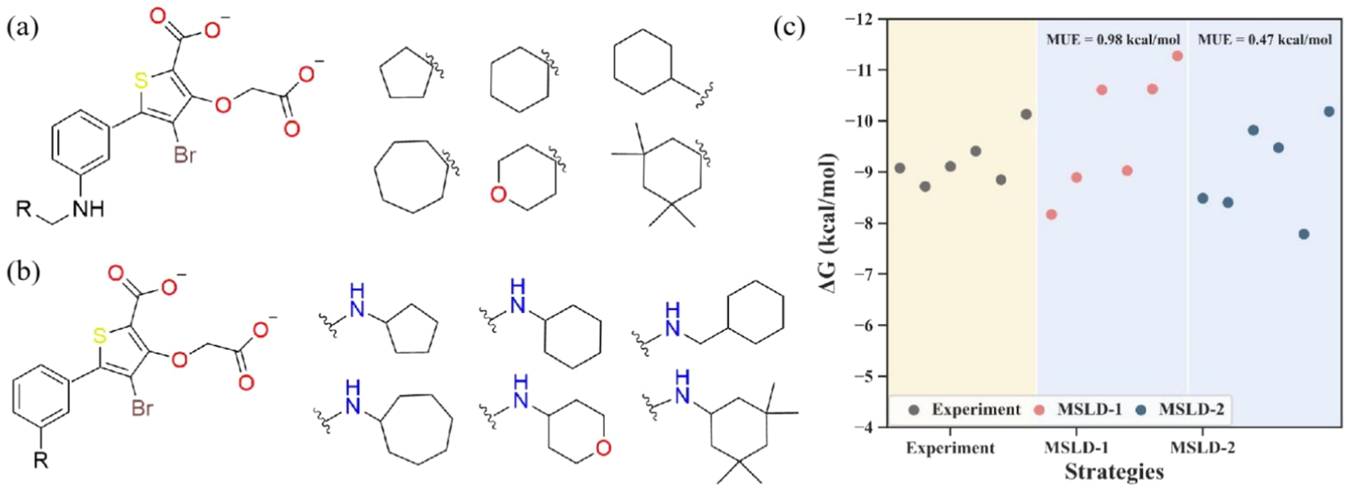

Figure 7. ΔG calculation results of the PTP1B-2 system. There are two ways to define the anchor atom: (a) the anchor atom is defined on the rings of perturbation groups and (b) the anchor atom is defined on the rings of the core. (c) ΔG calculation results. The gray scatter points represent the experimental binding affinities, the pink scatter points represent the calculation results obtained by defining the anchor atom on the rings of perturbation groups, and the dark blue scatter points represent the calculation results obtained by defining the anchor atom on the rings of the core. The MUEs for both calculation strategies are on the top of the corresponding area.

The topic of whether anchor atoms should be defined on the core's ring structure has grown in importance due to the numerous perturbing groups that involve rings.

The PTP1B-2 system contains 6 ligands modeled by defining the anchor atom on the rings of the perturbation groups and on the rings of the core separately, as shown in Figure 7a,7b. As shown in Figure 7c, the MUE calculated using the second definition (0.47 kcal/mol) is less than that calculated using the first definition (0.98 kcal/mol). However, when MSLD calculations are performed, the sampling of the latter is more challenging than that of the former. Figures S10 and S11 illustrate the potential landscapes between substituents within a site when performing MSLD simulations, from which the trend in the difficulty of sampling could be observed. That is to say, when the latter is used, difficulties with sampling may arise, thus requiring prolonged simulation times or parameter adjustments. This agrees with our previous understanding that the breaking and formation of rings during the perturbation would result in more complex and higher energy barriers. In brief, the first approach is a better choice as long as correcting errors induced by charge renormalization. Similar trends and conclusions may be demonstrated in the TYK2−3 system as well (Figures S12 and S14), further demonstrating that the former way converges faster but requires the correction of errors caused by charge renormalization.

Difference with Respect to DS Workflow. Our workflow incorporates the empirical strategies mentioned above along with specific considerations that are crucial for MSLD calculations. By utilizing default protocol and parameters, our workflow enables the acquisition of reliable results, even for inexperienced or exploratory participants. As was discussed in the previous paragraph, there was only a commercial DS workflow that existed before our own, and both workflows have the capability to reach considerable levels of prediction accuracy. Here, we conducted an analysis of the different options of our workflow and the DS workflow when handling various circumstances. It is important to note that the comparison provided below is primarily focused on how the underlying algorithms handle complex chemical issues rather than on workflow implementation advantages such as user-friendliness. We believe that by highlighting these differences in how our workflow and DS workflow handle numerous intricate chemical scenarios, users will be better equipped to identify the most suitable workflow for their specific requirements.

First, our workflow provides greater support for multisite perturbations compared to the DS workflow when addressing both single-site and multisite perturbations. According to the DS 2021 official documentation, it states that every noncore ligand should only differ from the core ligand at a single site.[23] Therefore, if perturbations occur at two or more sites, the direct identification of multisite ligands becomes impossible. In such cases, it is necessary to specify a single-site ligand for each site and then combine these single-site ligands to obtain the final required multisite ligand. A reminder here is that the DS benchmarking article did have 2-site and 3-site calculations, presumably done according to the above description.[22] While the preceding analysis indicates that the one-site approach is preferable for MSLD calculations, there exist scenarios where multisite perturbations are necessary, which may include the development of more advanced sampling methods or the exploration of novel perturbation modes.

Furthermore, our workflow encourages a broader spectrum of perturbation patterns. Panel b depicted in Figure 1 demonstrates the diverse perturbation patterns supported in our MCS search algorithm, particularly in the light of scaffold hopping (see MCL1−1 subset) and multiattachments of anchor atoms (see JNK1−1 subset, JNK1−2 subset, etc.). Thus, our workflow has the potential to facilitate scientific exploration of a greater variety of perturbation patterns. On the other hand, the perturbation patterns identified by the DS workflow tend to exhibit a relatively limited form. Moreover, the DS workflow's encapsulation level is too high for users to customize the perturbation patterns according to their specific needs.

Additionally, our workflow exhibits reasonable and appropriate performance when dealing with complicated ligand sets. This point has been clarified at the very beginning of the "Results and Discussion" Section.

Comparison with FEP. To further demonstrate the competitiveness of our MSLD workflow, FEP calculations of the thrombin system were performed and compared with the

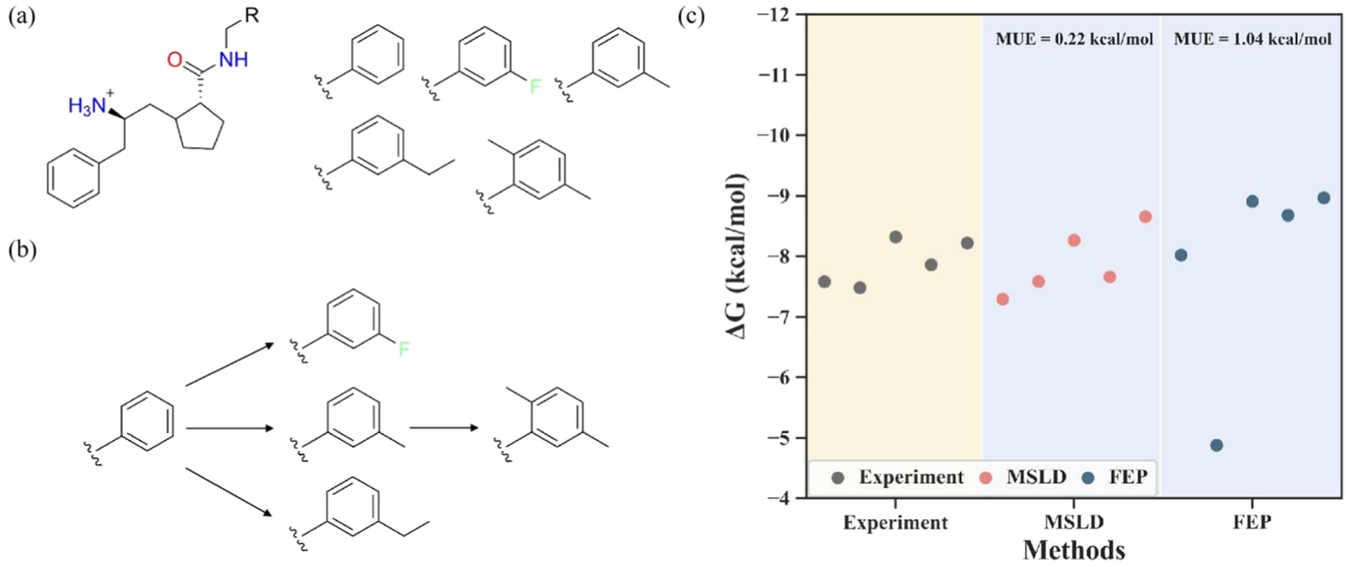

Figure 8. ΔG calculation results of the thrombin-1 system. (a) The perturbation pattern of MSLD. (b) The perturbation pattern of FEP, in which the perturbation groups are guaranteed to be consistent with those in MSLD instead of following the minimum perturbation path. (c) ΔG calculation results. The gray scatter points represent the experimental binding affinities; the pink scatter points represent the calculation results obtained by our MSLD workflow, and the dark blue scatter points represent the calculation results obtained by FEP. The MUEs for both calculation methods are on the top of the corresponding area.

MSLD results. The preparation of the proteins and ligands and the simulation system setup used in the FEP calculations were the same as those used in the MSLD calculations. The dual topology required by FEP was generated, and the perturbation part was identical to that in MSLD, as opposed to the minimum disturbance path typically employed. Figure 8b illustrates the network of the FEP calculations, where nodes represent different ligands and edges represent the corresponding alchemical transformations. The weight of the two change lines in the middle was set to 0.5. The FEP calculations were performed by the NAMD3 dynamics engine with 25 λ windows, each with 0.2 ns of equilibration and 0.6 ns of data collection for a total of 40 ns.[26,45] Similarly, three parallel simulations were run for the complex in solution and the ligands in solution separately. The relative binding free energies were postprocessed using the BAR estimator in the Alchemlyb package.[56] More details of the FEP preparation and simulation are given in the SI.

Figure 8c compares the ΔG values calculated by MSLD simulations (middle) and FEP simulations (right) with the experimental binding affinities (left). The MSLD calculations show a lower MUE (0.22 kcal/mol) than the FEP calculations, indicating a better agreement with the experimental data. The consistent trends in the experimental results, however, show that the FEP method captures the relative binding free energy differences between ligands. The computational efficiency of the MSLD and FEP calculations for this set of ligands was also evaluated. The MSLD calculations require a total of 300 ns of simulation and about 20 h of wall time on one GPU node. The FEP method required 3 times more simulation time (4 × 240 ns) and about 960 h of wall time on one GPU node. Therefore, the MSLD method achieved 1 order of magnitude in speedup over the FEP method while achieving comparable accuracy.

■ CONCLUSIONS

This study presents a customized workflow (along with a fully open-source package) for MSLD calculations based on the BLaDE simulation engine. This comprehensive and automated workflow provides reliable calculations (with high reproducibility) and requires minimal user intervention and input during the calculations. We hope that this carefully constructed workflow will facilitate broader adoption of this extremely useful molecular simulation technique. We have also made technical contributions in addition to putting together the workflow as well as a careful benchmark to further reveal the strengths and delicate aspects of MSLD. Specifically, our proposed (1) LIC-align algorithm and (2) modified MCS search algorithm are conducive to preparing the MSLD topologies for distinct ideal perturbation patterns. The encouraging results for 166 ligands across 8 proteins highlight the usefulness of our MSLD workflow for challenging but prevalent molecular modeling tasks, such as lead optimization in drug discovery. We have also pedantically analyzed the factors (including multisite versus single-site, grouping strategy, and how to define the anchor atoms) that may compromise MSLD calculations for different systems and proposed various coping strategies (to mitigate the undesirable effects) to strengthen the robustness of MSLD calculations. Our newly added preprocessing algorithms and protocols have significantly strengthened the robustness of a free energy calculation based on MSLD. Compared to the DS workflow, our workflow exhibits competitiveness in handling various tricky scenarios. In fact, substantial advantages in efficiency and comparable accuracy to the FEP calculations can now be routinely achieved for a broader range of complex systems, as illustrated by our case studies. This is the first comprehensive and open-source workflow based on the BLaDE simulation engine for MSLD calculations, and this package dramatically lowers the technical barriers to using MSLD-based free energy calculations for noncomputational scientists and researchers.

■ ASSOCIATED CONTENT

Data Availability Statement

The data sets used in this work, including BACE, CDK2, JNK1, MCL1, p38, PTP1B, thrombin, and TYK2, are provided by

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Wang et al.,[5] and they can be downloaded from https://pubs.acs.org/doi/10.1021/ja512751q. The raw data for all of the perturbations and the final calculated $\Delta\Delta G$ and ΔG for all of the ligands are given in the attached Excel file for download. The source codes of our comprehensive, open-source, and automated workflow are available in the GitHub repository: https://github.com/RenlingHu/WorkflowForMSLD.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.3c01154.

Details of grouping strategies, FEP preparations and calculations, and sample scripts; perturbation patterns for different systems (Figure S1−S8); correlation between the calculated and experimental ABFEs for the p38 systems (Figure S9); landscapes for the intratransformations of substituents within a site of PTP1B-2 (Figures S10 and S11); absolute binding free energy calculation results of the TYK2−3 system (Figure S12); and landscapes of the intratransformations of substituents within a site of TYK2 (Figures S13 and S14) (PDF)

Raw data of the MSLD calculations and FEP calculations (XLSX)

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Notes

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